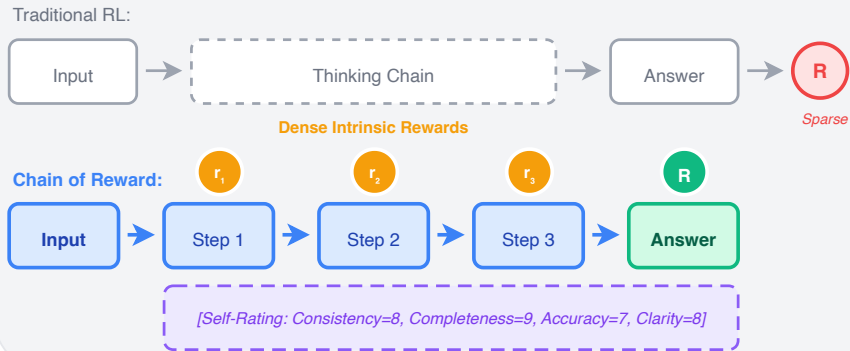


Chain of Reward (CoR) Framework

(a) Dense Rewards Along Reasoning Chain



(b) Four-Component Reward Decomposition

$$R(c) = R_{\text{ext}} + \lambda \cdot R_{\text{int}} + \mu \cdot R_{\text{improve}} + \nu \cdot R_{\text{converge}}$$

R_{ext}

External

Answer correctness
(sparse, binary)

R_{int}

Intrinsic

5D quality scores
+ self-rating calibration

R_{improve}

Improvement

$Q(c_{k+1}) - Q(c_k)$
iterative gain

R_{converge}

Convergence

Solution stability
across rounds

Dual-Coupled Evolutionary Dynamics

Better Policy $\rightarrow$ Higher-Quality Chains $\rightarrow$ Stronger Reward Signals $\rightarrow$ Better Policy $\odot$